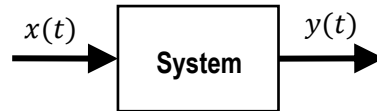


Worksheet 1 - Solution

Linearization

1) Consider a system with input $x(t)$ and output $y(t)$. The input-output relationship is provided as follows. In each case determine if the system is linear or nonlinear?

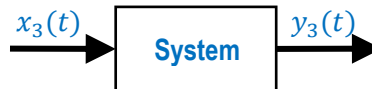


Linear systems must satisfy the **superposition** and **homogeneity** principle. Following are the steps to check it:

1- Find the system output to inputs $x_1(t)$ and $x_2(t)$



2- Find the system output to input $x_3(t) = \alpha x_1(t) + \beta x_2(t)$. x_3 is the linear combination of the inputs x_1 and x_2



3- Calculate the linear combination of the outputs y_1 and y_2 , which is $\alpha y_1(t) + \beta y_2(t)$

4- Compare the results of Step (2) and Step (3).

If $y_3(t) = \alpha y_1(t) + \beta y_2(t)$, the system is linear. Otherwise, the system is not linear.

a) $y(t) = 5x(t) + 7$

1- Find the system output to inputs $x_1(t)$ and $x_2(t)$

$$y_1(t) = 5x_1(t) + 7$$

$$y_2(t) = 5x_2(t) + 7$$

2- Find the system output to $x_3(t) = \alpha x_1(t) + \beta x_2(t)$

$$y_3(t) = 5x_3(t) + 7 = 5(\alpha x_1(t) + \beta x_2(t)) + 7 = 5\alpha x_1(t) + 5\beta x_2(t) + 7$$

3- Calculate the $\alpha y_1(t) + \beta y_2(t)$

$$\alpha y_1(t) + \beta y_2(t) = \alpha(5x_1(t) + 7) + \beta(5x_2(t) + 7) = 5\alpha x_1(t) + 5\beta x_2(t) + 7\alpha + 7\beta$$

4- Compare the results of Step (2) and Step (3)

Since $y_3(t) \neq \alpha y_1(t) + \beta y_2(t)$, **the system is not a linear system.**

b) $y(t) = 8x(t)$

1- Find the system output to inputs $x_1(t)$ and $x_2(t)$

$$y_1(t) = 8x_1(t)$$

$$y_2(t) = 8x_2(t)$$

2- Find the system output to $x_3(t) = \alpha x_1(t) + \beta x_2(t)$

$$y_3(t) = 8x_3(t) = 8(\alpha x_1(t) + \beta x_2(t)) = 8\alpha x_1(t) + 8\beta x_2(t)$$

3- Calculate the $\alpha y_1(t) + \beta y_2(t)$

$$\alpha y_1(t) + \beta y_2(t) = \alpha(8x_1(t)) + \beta(8x_2(t)) = 8\alpha x_1(t) + 8\beta x_2(t)$$

4- Compare the results of Step (2) and Step (3)

Since $y_3(t) = \alpha y_1(t) + \beta y_2(t)$, **the system is a linear system.**

c) $y(t) = tx(t)$

1- Find the system output to inputs $x_1(t)$ and $x_2(t)$

$$y_1(t) = tx_1(t)$$

$$y_2(t) = tx_2(t)$$

2- Find the system output to $x_3(t) = \alpha x_1(t) + \beta x_2(t)$

$$y_3(t) = tx_3(t) = t(\alpha x_1(t) + \beta x_2(t)) = \alpha tx_1(t) + \beta tx_2(t)$$

3- Calculate the $\alpha y_1(t) + \beta y_2(t)$

$$\alpha y_1(t) + \beta y_2(t) = \alpha(tx_1(t)) + \beta(tx_2(t)) = \alpha tx_1(t) + \beta tx_2(t)$$

4- Compare the results of Step (2) and Step (3)

Since $y_3(t) = \alpha y_1(t) + \beta y_2(t)$, **the system is a linear system.**

d) $y(t) = tx(t) + 1$

1- Find the system output to inputs $x_1(t)$ and $x_2(t)$

$$y_1(t) = tx_1(t) + 1$$

$$y_2(t) = tx_2(t) + 1$$

2- Find the system output to $x_3(t) = \alpha x_1(t) + \beta x_2(t)$

$$y_3(t) = tx_3(t) + 1 = t(\alpha x_1(t) + \beta x_2(t)) + 1 = \alpha tx_1(t) + \beta tx_2(t) + 1$$

3- Calculate the $\alpha y_1(t) + \beta y_2(t)$

$$\alpha y_1(t) + \beta y_2(t) = \alpha(tx_1(t) + 1) + \beta(tx_2(t) + 1) = \alpha tx_1(t) + \beta tx_2(t) + \alpha + \beta$$

4- Compare the results of Step (2) and Step (3)

Since $y_3(t) \neq \alpha y_1(t) + \beta y_2(t)$, **the system is not a linear system.**

e) $y(t) = e^{-x(t)}$

1- Find the system output to inputs $x_1(t)$ and $x_2(t)$

$$y_1(t) = e^{-x_1(t)}$$

$$y_2(t) = e^{-x_2(t)}$$

2- Find the system output to $x_3(t) = \alpha x_1(t) + \beta x_2(t)$

$$y_3(t) = e^{-x_3(t)} = e^{-(\alpha x_1(t) + \beta x_2(t))} = e^{-\alpha x_1(t)} e^{-\beta x_2(t)}$$

3- Calculate the $\alpha y_1(t) + \beta y_2(t)$

$$\alpha y_1(t) + \beta y_2(t) = \alpha(e^{-x_1(t)}) + \beta(e^{-x_2(t)}) = \alpha e^{-x_1(t)} + \beta e^{-x_2(t)}$$

4- Compare the results of Step (2) and Step (3)

Since $y_3(t) \neq \alpha y_1(t) + \beta y_2(t)$, **the system is not a linear system.**

2) Assume a nonlinear resistor whose voltage-current relationship is defined by,

$$i = 2e^{0.1v}$$

where i and v are the resistor current and voltage, respectively.

Find a linear relationship for voltage-current around the operation point of $v_0 = 20V$.

We do the linearization using Taylor Series approximation around the operation point of $v_0 = 20V$.

Here, current i is a function of voltage v . The current value at operation point is:

$$i_0 = 2e^{0.1v_0} = 2e^{0.1(20)} = 2e^2 = 14.78 \text{ A}$$

The linear approximation around the operation point will have the following general form:

$$i - i_0 = m_A(v - v_0) \quad \rightarrow \quad \delta i = m_A \delta v$$

where m_A is the slope of the tangent line to the curve of $i = 2e^{0.1v}$ at the operation point:

$$m_A = \left. \frac{di}{dv} \right|_{v=v_0} = 0.2e^{0.1v}|_{v=20} = 0.2e^2 = 1.478$$

The linear voltage-current relationship for small excursions of voltage around the operation point of $v_0 = 20V$:

$$\delta i = 1.478 \delta v \quad \text{or} \quad \delta v = 0.677 \delta i$$

3) Assume a nonlinear resistor whose voltage-current relationship is defined by,

$$i = 2v^3$$

where i and v are the resistor current and voltage, respectively.

Find a linear relationship for voltage-current around the operation point of $v_0 = 2V$.

We do the linearization using Taylor Series approximation around the operation point of $v_0 = 2V$.

Here, current i is a function of voltage v . The current value at operation point is:

$$i_0 = 2v_0^3 = 2(2)^3 = 16 \text{ A}$$

The linear approximation around the operation point has the following general form:

$$i - i_0 = m_A(v - v_0) \quad \rightarrow \quad \delta i = m_A \delta v$$

where m_A is the slope of the tangent line to the curve of $i = 2v^3$ at the operation point:

$$m_A = \left. \frac{di}{dv} \right|_{v=v_0} = 6v^2|_{v=2} = 6(4) = 24$$

The linear voltage-current relationship for small excursions of voltage around the operation point of $v_0 = 2V$:

$$\delta i = 24 \delta v \quad \text{or} \quad \delta v = 0.042 \delta i$$

4) Determine which differential equation is linear time-invariant?

a) $\ddot{y}(t) + 5y^2(t) = 0$

Due to the $y^2(t)$ term, the equation is nonlinear.

b) $10t\ddot{y}(t) + 5\dot{y}(t) + y(t) = 0$

Due to the $10t$ term, the equation is time-varying.

c) $\dot{y}(t) + 6y(t) = 0$

The equation is linear and time-invariant.

d) $\sin(t)\ddot{y}(t) + 5\dot{y}(t) + 2y(t) = 0$

Due to the $\sin t$ term, the equation is time-varying.

e) $8\ddot{y}(t) - 2t^2\dot{y}(t) + 5\ln(y(t)) = 0$

Due to the $-2t^2$ term, the equation is time-varying and due to the $\ln(y(t))$ term, it is nonlinear.

5) Consider the following differential equation:

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = x$$

where x is the input and is a function of output y . If $x = \sin y$, linearize the differential equation for small excursions around:

a) $y = 0$

Small excursion about the operation point of $y_0 = 0$ is $\rightarrow \delta y = y - y_0 = y - 0 = y$

Set $y = \delta y$ into the equation

$$\frac{d^2\delta y}{dt^2} + 3\frac{d\delta y}{dt} + 2\delta y = \sin \delta y$$

Linearized the term $\sin \delta y$ around $y_0 = 0$ by Taylor series approximation of: $f(y) = f(y_0) + m_A(y - y_0)$

$$f(y) = \sin \delta y \rightarrow f(y_0) = \sin 0 = 0$$

$$\frac{df(y)}{dy} = \cos y \rightarrow m_A = \left. \frac{df(y)}{dy} \right|_{y=y_0} = \cos 0 = 1 \Rightarrow f(y) = f(y_0) + m_A(y - y_0) = 0 + 1(\underbrace{y - 0}_{\delta y}) = \delta y$$

Thus, the linear approximation of the differential equation around point $y_0 = 0$ is:

$$\frac{d^2\delta y}{dt^2} + 3\frac{d\delta y}{dt} + 2\delta y = \delta y \rightarrow \boxed{\frac{d^2\delta y}{dt^2} + 3\frac{d\delta y}{dt} + \delta y = 0}$$

Solve for δy and find $y = \delta y + y_0 = \delta y$.

b) $y = \pi$

Small excursion about the operation point of $y_0 = \pi$ is $\rightarrow \delta y = y - y_0 = y - \pi$

Set $y = \delta y + \pi$ into the equation

$$\frac{d^2 \delta y}{dt^2} + 3 \frac{d\delta y}{dt} + 2(\delta y + \pi) = \sin(\delta y + \pi)$$

Linearized the term $\sin \delta y$ around $y_0 = \pi$ by Taylor series approximation of: $f(y) = f(y_0) + m_A(y - y_0)$

$$f(y) = \sin(\delta y + \pi) \rightarrow f(y_0) = \sin \pi = 0$$

$$\frac{df(y)}{dy} = \cos y \rightarrow m_A = \left. \frac{df(y)}{dy} \right|_{y=y_0} = \cos \pi = -1 \quad \Rightarrow \quad f(y) = f(y_0) + m_A(y - y_0) = 0 - 1(\underbrace{y - \pi}_{\delta y}) = -\delta y$$

Thus, the linear approximation of the differential equation around point $y_0 = \pi$ is:

$$\frac{d^2 \delta y}{dt^2} + 3 \frac{d\delta y}{dt} + 2(\delta y + \pi) = -\delta y \rightarrow \boxed{\frac{d^2 \delta y}{dt^2} + 3 \frac{d\delta y}{dt} + 3\delta y = -2\pi}$$

Solve for δy and find $y = \delta y + y_0 = \delta y + \pi$.

6) Linearized the following differential equation for y near 0.

$$\frac{d^3 y}{dt^3} + 10 \frac{d^2 y}{dt^2} + 31 \frac{dy}{dt} + 30y = 3e^{-5y}$$

Small excursion about the operation point of $y_0 = 0$ is $\rightarrow \delta y = y - y_0 = y$

Set $y = \delta y$ into the equation

$$\frac{d^3 \delta y}{dt^3} + 10 \frac{d^2 \delta y}{dt^2} + 31 \frac{d\delta y}{dt} + 30\delta y = 3e^{-5\delta y}$$

Linearized the term $3e^{-5\delta y}$ around $y_0 = 0$ by Taylor series approximation of: $f(y) = f(y_0) + m_A(y - y_0)$

$$f(y) = 3e^{-5\delta y} \rightarrow f(y_0) = 3e^0 = 3$$

$$\frac{df(y)}{dy} = -15e^{-5y} \rightarrow m_A = \left. \frac{df(y)}{dy} \right|_{y=y_0} = -15e^0 = -15 \quad \Rightarrow \quad f(y) = f(y_0) + m_A(y - y_0) = 3 - 15(\underbrace{y - 0}_{\delta y}) = 3 - 15\delta y$$

Thus, the linear approximation of the differential equation around point $y_0 = \pi$ is:

$$\frac{d^3 \delta y}{dt^3} + 10 \frac{d^2 \delta y}{dt^2} + 31 \frac{d\delta y}{dt} + 30\delta y = 3 - 15\delta y \rightarrow \boxed{\frac{d^3 \delta y}{dt^3} + 10 \frac{d^2 \delta y}{dt^2} + 31 \frac{d\delta y}{dt} + 45\delta y = 3}$$

Solve for δy and find $y = \delta y + y_0 = \delta y$.

7) Many systems are *piecewise* linear. A system with amplifier saturation is one such example. Given the differential equation

$$\ddot{x}(t) + 17\dot{x}(t) + 50x = f(x)$$

Assume that $f(x)$ is as shown in the figure.

Write the differential equation for each of the following ranges of x :

a) $-\infty < x < -3$

For $-\infty < x < -3$, the function is described as $f(x) = -6$.

Thus, the differential equation is:

$$\ddot{x}(t) + 17\dot{x}(t) + 50x = -6$$

b) $-3 < x < 3$

For $-3 < x < 3$, the function is described as $f(x) = 2x$.

Thus, the differential equation is:

$$\ddot{x}(t) + 17\dot{x}(t) + 50x = 2x \quad \rightarrow \quad \ddot{x}(t) + 17\dot{x}(t) + 48x = 0$$

c) $3 < x < \infty$

For $3 < x < \infty$, the function is described as $f(x) = 6$.

Thus, the differential equation is:

$$\ddot{x}(t) + 17\dot{x}(t) + 50x = 6$$

